Experiment results - Market-1501, body appearance only

Measured 2026-10-25 / 2026-11-03 on the development laptop. Protocol: EVALUATION_PROTOCOL.md. Raw runs: `docs/results/market1501/` *and* `docs/results/market1501_finetuned/`*. Regenerate this page with* `python scripts/build_experiment_results.py`.

Read this first. These are body-appearance numbers on a public benchmark, with the project's own identity-disjoint protocol on a CPU-sized subset (no official `query/` folder was available, so they are *not* comparable with published Market-1501 results). Face + body cannot be benchmarked on Market-1501 (the images are too small for faces). Known limits: LIMITATIONS.md.

1. Setup in one table

Dataset	Market-1501, 26,051 labelled images / 1,501 identities / 6 cameras (pool of train + test folders)
Per split (seed 0)	60 enrolled + 60 stranger people; 3 gallery photos per enrolled person; 290 / 290 known queries and 458 / 465 stranger queries (calibration / test)
Seeds	0, 1, 2 (each re-draws who is enrolled and who is a stranger); the fine-tuning experiment uses seed 0 only
Model	OSNet x1.0 (multi-source) - official osnet_x1_0_ms_d_c.pt (verified: MODEL_VERIFICATION.md)
Hardware / software	Intel Core i5-7300U (2 cores / 4 threads), 7.9 GB RAM, no GPU; Python 3.14.4, torch 2.14.1+cpu, numpy 2.5.2
Speed (measured)	embedding one crop: p50 196 ms, p95 235 ms; batch of 16: 5.3 crops/s; retrieval per query (180-image gallery): 0.017 ms; model load 0.62 s; peak memory 613 MB. **Not real-time video**

2. FAR at a fixed recall - the PS-1 objective (3 seeds, TEST people)

For each target recall (DIR@1 = enrolled people accepted **and** correctly named) the threshold is chosen on the **calibration** people to give the lowest FAR that still reaches the target, frozen, and scored once on the disjoint **test** people. FAR = strangers for whom *any* enrolled person was named. Entries are mean ± standard deviation over the seeds.

Target DIR@1 (calibration)	Variant	Target reached on calibration	Test DIR@1 (recall)	Test FAR	ROC-AUC	Rank-1	mAP	Threshold
50%	Nearest neighbour (no rejection)	0 of 3 seeds	83.6 % ± 1.8	100.0 % ± 0.0	n/a	83.6 % ± 1.8	77.8 % ± 1.4	n/a
50%	+ Calibrated threshold	3 of 3 seeds	53.3 % ± 3.5	6.4 % ± 3.7	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.775 ± 0.008
50%	+ Top-1/top-2 margin rule	3 of 3 seeds	50.5 % ± 2.3	6.2 % ± 2.9	0.770 ± 0.036	83.6 % ± 1.8	77.8 % ± 1.4	0.749 ± 0.027
50%	+ Calibrated confidence (instead of margin)	3 of 3 seeds	54.7 % ± 2.5	7.0 % ± 1.4	0.847 ± 0.017	83.6 % ± 1.8	77.8 % ± 1.4	0.500 ± 0.043
50%	+ Average over a person's photos (instead of the best photo)	3 of 3 seeds	53.7 % ± 9.4	5.4 % ± 3.9	0.861 ± 0.026	87.8 % ± 3.8	77.8 % ± 1.4	0.725 ± 0.007
50%	+ Flip test-time augmentation	3 of 3 seeds	52.5 % ± 0.9	7.6 % ± 3.5	0.760 ± 0.028	83.4 % ± 1.5	77.9 % ± 1.3	0.733 ± 0.029

Target DIR@1 (calibration)	Variant	Target reached on calibration	Test DIR@1 (recall)	Test FAR	ROC-AUC	Rank-1	mAP	Threshold
60%	Nearest neighbour (no rejection)	0 of 3 seeds	83.6 % ± 1.8	100.0 % ± 0.0	n/a	83.6 % ± 1.8	77.8 % ± 1.4	n/a
60%	+ Calibrated threshold	3 of 3 seeds	64.3 % ± 2.2	12.9 % ± 4.9	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.749 ± 0.006
60%	+ Top-1/top-2 margin rule	3 of 3 seeds	60.5 % ± 0.2	13.8 % ± 4.2	0.779 ± 0.020	83.6 % ± 1.8	77.8 % ± 1.4	0.711 ± 0.014
60%	+ Calibrated confidence (instead of margin)	3 of 3 seeds	61.7 % ± 3.3	11.4 % ± 2.5	0.847 ± 0.017	83.6 % ± 1.8	77.8 % ± 1.4	0.352 ± 0.047
60%	+ Average over a person's photos (instead of the best photo)	3 of 3 seeds	62.9 % ± 8.7	12.2 % ± 8.0	0.861 ± 0.026	87.8 % ± 3.8	77.8 % ± 1.4	0.703 ± 0.010
60%	+ Flip test-time augmentation	3 of 3 seeds	60.2 % ± 1.1	13.9 % ± 4.8	0.776 ± 0.020	83.4 % ± 1.5	77.9 % ± 1.3	0.710 ± 0.023
70%	Nearest neighbour (no rejection)	0 of 3 seeds	83.6 % ± 1.8	100.0 % ± 0.0	n/a	83.6 % ± 1.8	77.8 % ± 1.4	n/a
70%	+ Calibrated threshold	3 of 3 seeds	72.7 % ± 2.2	28.6 % ± 8.0	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.715 ± 0.008
70%	+ Top-1/top-2 margin rule	3 of 3 seeds	71.4 % ± 2.1	25.0 % ± 4.7	0.827 ± 0.048	83.6 % ± 1.8	77.8 % ± 1.4	0.710 ± 0.011
70%	+ Calibrated confidence (instead of margin)	3 of 3 seeds	73.3 % ± 1.0	25.5 % ± 4.8	0.847 ± 0.017	83.6 % ± 1.8	77.8 % ± 1.4	0.179 ± 0.034
70%	+ Average over a person's photos (instead of the best photo)	3 of 3 seeds	70.5 % ± 5.0	21.4 % ± 7.9	0.861 ± 0.026	87.8 % ± 3.8	77.8 % ± 1.4	0.680 ± 0.009
70%	+ Flip test-time augmentation	3 of 3 seeds	73.0 % ± 0.7	28.8 % ± 4.6	0.814 ± 0.025	83.4 % ± 1.5	77.9 % ± 1.3	0.696 ± 0.009
80%	Nearest neighbour (no rejection)	0 of 3 seeds	83.6 % ± 1.8	100.0 % ± 0.0	n/a	83.6 % ± 1.8	77.8 % ± 1.4	n/a
80%	+ Calibrated threshold	2 of 3 seeds	81.5 % ± 1.4	79.7 % ± 19.9	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.615 ± 0.073
80%	+ Top-1/top-2 margin rule	2 of 3 seeds	80.9 % ± 2.2	76.0 % ± 21.2	0.836 ± 0.039	83.6 % ± 1.8	77.8 % ± 1.4	0.599 ± 0.060
80%	+ Calibrated confidence (instead of margin)	2 of 3 seeds	81.7 % ± 1.2	77.9 % ± 21.3	0.847 ± 0.017	83.6 % ± 1.8	77.8 % ± 1.4	0.048 ± 0.036
80%	+ Average over a person's photos (instead of the best photo)	3 of 3 seeds	84.1 % ± 1.6	59.5 % ± 11.2	0.861 ± 0.026	87.8 % ± 3.8	77.8 % ± 1.4	0.620 ± 0.019
80%	+ Flip test-time augmentation	2 of 3 seeds	82.0 % ± 1.7	80.1 % ± 17.2	0.838 ± 0.037	83.4 % ± 1.5	77.9 % ± 1.3	0.584 ± 0.050
90%	Nearest neighbour (no rejection)	0 of 3 seeds	83.6 % ± 1.8	100.0 % ± 0.0	n/a	83.6 % ± 1.8	77.8 % ± 1.4	n/a
90%	+ Calibrated threshold	0 of 3 seeds	83.6 % ± 1.8	99.6 % ± 0.3	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.550 ± 0.016

Target DIR@1 (calibration)	Variant	Target reached on calibration	Test DIR@1 (recall)	Test FAR	ROC-AUC	Rank-1	mAP	Threshold
90%	+ Top-1/top-2 margin rule	0 of 3 seeds	83.6 % ± 1.8	99.6 % ± 0.3	0.850 ± 0.027	83.6 % ± 1.8	77.8 % ± 1.4	0.550 ± 0.016
90%	+ Calibrated confidence (instead of margin)	0 of 3 seeds	83.6 % ± 1.8	99.6 % ± 0.3	0.847 ± 0.017	83.6 % ± 1.8	77.8 % ± 1.4	0.013 ± 0.006
90%	+ Average over a person's photos (instead of the best photo)	0 of 3 seeds	87.8 % ± 3.8	99.7 % ± 0.3	0.861 ± 0.026	87.8 % ± 3.8	77.8 % ± 1.4	0.496 ± 0.020
90%	+ Flip test-time augmentation	0 of 3 seeds	83.4 % ± 1.5	99.5 % ± 0.5	0.849 ± 0.027	83.4 % ± 1.5	77.9 % ± 1.3	0.549 ± 0.019

What the table says

- **Embedding quality is good:** closed-set Rank-1 83.6 % ± 1.8, mAP 77.8 % ± 1.4 (best photo per person), ROC-AUC 0.850 ± 0.027 for telling enrolled people from strangers.
- **The open-set price is high.** With the plain threshold rule the test FAR is **6.4 %** at 50 % recall, **12.9 %** at 60 %, **28.6 %** at 70 %, and about **80 %** at 80 % recall (the 80 % target was reached on calibration for only 2 of 3 seeds). **A 90 % recall target is unreachable** - the best possible DIR@1 is the Rank-1 of 83.6 % - and the best-effort threshold accepts 99.6 % of strangers. Body appearance alone cannot give a low FAR at high recall; this is the "low inter-class variance" problem, measured.
- **Negative result - the top-1/top-2 margin rule did not help:** at 70 % recall FAR 25.0 % ± 4.7 vs 28.6 % ± 8.0 for the plain threshold (inside the noise) and it lowered recall and ROC-AUC. **Calibrated confidence** was equivalent (25.5 % ± 4.8). **Flip augmentation** gave nothing (28.8 % ± 4.6).
- **Averaging a person's photos** (instead of taking the best photo) was the best *free* change: FAR 21.4 % ± 7.9 at 70 % recall and 59.5 % ± 11.2 at 80 % (reached on all 3 seeds), Rank-1 87.8 % ± 3.8 - but the standard deviations are large and the recall actually reached differs between variants (70.5 % vs 72.7 %), so treat it as *promising, not proven*.
- **Threshold-only rejection has a structural weakness:** a stranger only has to resemble *one* of 60 enrolled people (180 photos) to be accepted, so FAR grows quickly as the gallery grows. This is why the product uses the face first.

3. Does metric-learning fine-tuning help? (seed 0 only, TEST people)

The last stage of the network was fine-tuned for 3 epochs × 30 batches (16 people × 4 photos) with **batch-hard triplet loss** (cosine distance, margin 0.3, hard-negative mining inside each batch) on the **590 development people only** (8,759 images, 2,650 s on this CPU). The evaluation refuses a checkpoint if any person it was trained on appears in the calibration or test split (a leakage guard with its own test); none did. Checkpoint: `backend/weights/finetuned/market_seed0_head.pt` (git-ignored). Training seed: 0; **one seed, one training run - no confidence statement is possible beyond "this run"**.

Target DIR@1 (calibration)	Variant	Target reached on calibration	Test DIR@1 (recall)	Test FAR	ROC-AUC	Rank-1	mAP	Threshold
50%	+ Calibrated threshold	1 of 1 seeds	55.5 % ± 0.0	3.9 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.780 ± 0.000
50%	+ Top-1/top-2 margin rule	1 of 1 seeds	53.1 % ± 0.0	3.4 % ± 0.0	0.797 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.760 ± 0.000
50%	+ Triplet fine-tuned backbone	1 of 1 seeds	60.0 % ± 0.0	6.0 % ± 0.0	0.811 ± 0.000	89.7 % ± 0.0	87.3 % ± 0.0	0.795 ± 0.000
60%	+ Calibrated threshold	1 of 1 seeds	64.8 % ± 0.0	8.4 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.752 ± 0.000
60%	+ Top-1/top-2 margin rule	1 of 1 seeds	60.7 % ± 0.0	9.7 % ± 0.0	0.797 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.712 ± 0.000

Target DIR@1 (calibration)	Variant	Target reached on calibration	Test DIR@1 (recall)	Test FAR	ROC-AUC	Rank-1	mAP	Threshold
60%	+ Triplet fine-tuned backbone	1 of 1 seeds	64.1 % ± 0.0	6.0 % ± 0.0	0.901 ± 0.000	89.7 % ± 0.0	87.3 % ± 0.0	0.806 ± 0.000
70%	+ Calibrated threshold	1 of 1 seeds	73.8 % ± 0.0	22.2 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.719 ± 0.000
70%	+ Top-1/top-2 margin rule	1 of 1 seeds	73.8 % ± 0.0	22.2 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.719 ± 0.000
70%	+ Triplet fine-tuned backbone	1 of 1 seeds	72.1 % ± 0.0	11.4 % ± 0.0	0.901 ± 0.000	89.7 % ± 0.0	87.3 % ± 0.0	0.779 ± 0.000
80%	+ Calibrated threshold	0 of 1 seeds	83.1 % ± 0.0	100.0 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.532 ± 0.000
80%	+ Top-1/top-2 margin rule	0 of 1 seeds	83.1 % ± 0.0	100.0 % ± 0.0	0.881 ± 0.000	83.1 % ± 0.0	76.3 % ± 0.0	0.532 ± 0.000
80%	+ Triplet fine-tuned backbone	1 of 1 seeds	87.9 % ± 0.0	58.5 % ± 0.0	0.901 ± 0.000	89.7 % ± 0.0	87.3 % ± 0.0	0.642 ± 0.000

(The `+ Calibrated threshold` and `+ Top-1/top-2 margin rule` rows are the **pretrained** model on the same seed-0 people; `+ Triplet fine-tuned backbone` is the fine-tuned one.)

- Closed-set quality rose from **Rank-1 83.1 % → 89.7 %** and **mAP 76.3 % → 87.3 %**.
- At **70 % recall** the false-accept rate fell from **22.2 % to 11.4 %**; at **80 %** the target went from **unreachable (FAR 100 %) to reachable with FAR 58.5 %**.
- At the **lowest recalls the result is mixed** (50 %: FAR 3.4 % → 6.0 %, i.e. not better; 60 %: 9.7 % → 6.0 %). With 60 stranger people the FAR resolution is coarse, and this is a single seed.
- The fine-tuned checkpoint is **not** the product default: it is trained on Market-1501 people and would need re-validation on any other camera. Use it via `REID_EMBEDDING_CHECKPOINT`.

4. What was not run

`NOT_RUN`: fine-tuning for seeds 1 and 2 (about 45 minutes each on this CPU); full-backbone fine-tuning (`--trainable all`); any experiment on a second dataset or on same-clothing ground truth; face + body on pedestrian data.